Core Formalism of Structural Selection

A Reader's Guide to the MAAT Series

Defect Geometry, Response Closure, and Observable Layers

MAAT Structural Selection Series — Introductory Note

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Abstract

This note introduces the mathematical core of the structural-selection framework before the reader encounters the individual papers. It is intended to be read first, as a map and reference key for the series.

The framework is formulated as an effective structural-selection layer defined over admissible configuration spaces. Primitive defect sectors, support mappings, maximum-entropy selection functionals, covariance-driven response closure, emergent robustness relations, and observable projection layers are unified into a single formal architecture. The primary test field for the observable projection layer is cosmological redshift space, where the framework makes contact with expansion history $H(z)$ and growth rate $f\sigma_8(z)$.

The central derivation pipeline is:

$$\text{defects} \rightarrow \text{support geometry} \rightarrow \text{selection functional}$$
$$\rightarrow \text{response closure} \rightarrow \text{observable projections.}$$

This paper does not propose a completed fundamental theory, ultraviolet completion, or precision cosmological model. Its purpose is formal consolidation: to define the canonical variables, mappings, and hierarchical structure underlying the existing series, and to establish a reproducible mathematical basis for future field-theoretic, optimisation, and cosmological extensions.

Contents

1 Introduction **2**

2 Scope and Status **2**

3 Primitive Configuration Structure **3**

4 Support Geometry **3**

5 Structural Selection Functional **4**

6 Covariance-Response Closure **4**

7 Emergent Robustness Closure **5**

8 Critical Coherence Index **5**

9 Dynamical Extension 6

10 Observable Projection Layer 7

11 Architecture Diagram 7

12 Mathematical Status 8

13 Open Problems 8

14 Conclusion 9

1 Introduction

The structural-selection programme originated from a fundamental question:

Why do some dynamically admissible configurations persist, while others rapidly decay or fail to appear?

The individual papers introduce the Critical Coherence Index (CCI), structural free-energy functionals [9, 10, 11], covariance-response closures [13], and observable projection layers [15, 16]. These developments are distributed across multiple papers and intermediate formulations.

How to use this note. Read this note before the series to understand the notation, architecture, and scope. Each section corresponds to a layer of the framework. Cross-references point to the papers where each layer is developed in detail. This note is not itself a research paper — it is a mathematical map of the series.

This note defines the canonical mathematical structure and clarifies the hierarchy between:

1. primitive defect sectors (d_H, d_B, d_S, d_V),
2. bounded support mappings $\Gamma_a = 1/(1 + d_a)$,
3. structural selection functionals $\mathcal{F}_{\text{struct}}[X]$,
4. covariance-response closures for λ_a,
5. emergent robustness relations $(R_{\text{resp}}, R_{\text{rob}})$,
6. observable projection layers $(C_{\text{proj}}(z), z_{\text{tr}})$.

The framework is an effective structural-selection layer, not a completed fundamental theory.

2 Scope and Status

This framework is:

- an effective structural-selection layer,
- a diagnostic and response-based formalism,
- a reproducible phenomenological architecture,
- a toy-level observable projection framework.

This framework is not:

- a UV-complete theory,
- a validated modification of General Relativity [8],
- a precision cosmological fit or Boltzmann-code calculation,
- a derivation of the Standard Model spectrum,
- a proof of modified gravity or dark energy.

Observable layers should be interpreted as diagnostic and phenomenological projections, not as established physical observables.

3 Primitive Configuration Structure

Let $X \in \mathcal{M}$ denote an admissible configuration in configuration space $\mathcal{M}$.

Primitive defect vector.

$$d(X) = (d_H,\, d_B,\, d_S,\, d_V), \qquad d_a(X) \geq 0\ \forall a. \tag{1}$$

Smaller defect values correspond to stronger structural support. All defects are non-negative; zero defect means a perfectly satisfied constraint.

Table 1: Physical interpretation of the four primitive defect sectors.

Sector	Symbol	Physical meaning
Harmony	d_H	Equation-of-motion residual; zero when field equations are satisfied exactly
Balance	d_B	Divergence of stress-energy tensor; zero when conservation holds
Activity / Structure	d_S	Target-normalised activity defect; $((A - A_*)/\sigma_A)^2$
Connectedness	d_V	$1 - \tilde{I}_{\mathrm{nn}}$, normalised nearest-neighbour mutual information defect

The normalisation $\tilde{I}_{\mathrm{nn}} \in [0, 1]$ ensures $d_V \geq 0$ always. The activity sector d_S selects the structurally active intermediate regime, not the highest or lowest activity [18].

4 Support Geometry

Primitive defects are mapped to bounded support variables:

$$\Gamma_a(X) = \frac{1}{1 + d_a(X)}, \qquad 0 < \Gamma_a \leq 1. \tag{2}$$

This mapping guarantees:

Property	Consequence
Positivity	$\Gamma_a > 0$ always
Boundedness	$\Gamma_a \leq 1$
Monotonicity	increasing with structural health
Normalization	dimensionless, regime-independent

The support vector is $\Gamma(X) = (\Gamma_H, \Gamma_B, \Gamma_S, \Gamma_V)$. By construction, $\Gamma_a \to 1$ when the corresponding structural constraint is perfectly satisfied, and $\Gamma_a \to 0$ in the limit of critical defect.

5 Structural Selection Functional

Selection functional (v1.2.1 form).

$$\mathcal{F}_{\mathrm{MAAT}}[X] = -\sum_{a\in\{H,B,S,V\}} \lambda_a \log(\varepsilon + \Gamma_a[X]) - \lambda_{\mathrm{cl}} \log(\varepsilon + R_{\mathrm{rob}}[X]), \tag{3}$$

$$P[X] \propto \exp(-\mathcal{F}_{\mathrm{MAAT}}[X])\,. \tag{4}$$

Symbol reference:

Symbol	Range	Role
$\lambda_a^{(+)} = \max(0, \lambda_a^{\mathrm{raw}})$	≥ 0	penalty weight for primary sector a
$\lambda_{\mathrm{cl}} = \sum_a w_a \lambda_a$	≥ 0	closure coefficient for R_{rob}; derived, not primary
ε	$\ll 1$	regulariser preventing $\log(0)$

This defines structural selection as an exponential tilting over admissible configuration space, derived from maximum-entropy principles [1, 2, 3]. The MAAT functional is a specific closure realisation of the general structural functional:

$$\mathcal{F}_{\mathrm{MAAT}} \equiv \mathcal{F}_{\mathrm{struct}}\big|_{\text{v1.2.1 closure}}.$$

6 Covariance-Response Closure

The structural weights are not arbitrary constants. They are generated from the defect covariance geometry [13, 5, 6]:

$$\lambda = \left(C + \eta\,\frac{\mathrm{tr}(C)}{A}\,I\right)^{-1} (\langle d\rangle_{\mu_0} - d^*)\,, \tag{5}$$

Symbol	Meaning
$C = \mathrm{Cov}_{\mu_0}(d_a, d_b)$	defect covariance matrix under reference measure μ_0
$\langle d\rangle_{\mu_0}$	mean defect vector under the reference measure
d^*	target defect vector (desired structural state)
η	ridge regularisation parameter
A	number of sectors ($A = 4$)

The raw weights λ_a^{raw} may take any sign. For the penalty functional (3), only $\lambda_a^{(+)} = \max(0, \lambda_a^{\text{raw}})$ are used. For response-share normalisation, absolute values $\pi_a = |\lambda_a^{\text{raw}}| / \sum_b |\lambda_b^{\text{raw}}|$ apply.

The weights therefore encode the response pressure required to shift the ensemble from $\langle d \rangle_{\mu_0}$ toward d^* given the defect covariance geometry.

7 Emergent Robustness Closure

v1.2.1 robustness closure.

$$R_{\text{resp}} = (H \cdot B \cdot V)^{1/3}, \tag{6}$$

$$R_{\text{rob}} = \min\Big(R_{\text{resp}},\ (H \cdot B \cdot S \cdot V)^{1/4}\Big). \tag{7}$$

Since $R_{\text{rob}} \leq R_{\text{resp}}$ by construction, robustness is the equal-or-stronger stability constraint; equality holds when $S \geq (HBV)^{1/12}$.
No independent fifth primitive sector is introduced.

Interpretation. R_{resp} is the implicit structural respect — the degree to which the configuration adheres to its own constraints. R_{rob} is the emergent robustness — the capacity to withstand perturbation. The distinction was formalised in Papers 36–37 and 41 [17, 18].

Quantity	Formula	Meaning
R_{resp}	$(HBV)^{1/3}$	constraint adherence
R_{rob}	$\min(R_{\text{resp}}, (HBSV)^{1/4})$	stability under perturbation
λ_{cl}	$\sum_a w_a \lambda_a$	closure coefficient (derived)
Stability	$= R_{\text{rob}}$	selection stability diagnostic

8 Critical Coherence Index

The CCI is interpreted hierarchically across three levels.

Formal definition.

$$\mathcal{C}_{\text{CCI}} = \frac{\Gamma_{\text{inst}}\, \Gamma_{\text{prod}}\, (1 + \kappa U_{\text{struct}})}{\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon}. \tag{8}$$

Physical proxy (nonlinear field-theory benchmarks):

$$\mathcal{C}_{\text{CCI}} \approx \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}} + \varepsilon}. \tag{9}$$

Empirical scaling (ensemble fits):

$$\mathcal{C}_{\text{CCI}} \sim \frac{\dot{S}_{\text{cg}}^{(+)}}{I_{\text{nn}}^{\alpha}}, \qquad \alpha \approx 2.5\ (1\text{D}),\ 3.0\ (2\text{D}). \tag{10}$$

Regime	I_{nn}	$\dot{S}_{\text{cg}}^{(+)}$
Ordered	high	low
Critical	moderate	moderate
Chaotic	low	high

Symbol	Meaning
$\dot{S}_{\mathrm{cg}}^{(+)}$	positive part of the coarse-grained entropy production rate; measures structural activity
I_{nn}	nearest-neighbour mutual information; measures spatial coherence and coupling ($\tilde{I}_{\mathrm{nn}} = I_{\mathrm{nn}}/\log\lvert\mathcal{X}\rvert \in [0,1]$)
α	empirical scaling exponent; $\alpha \approx 2.5$ (1D), 3.0 (2D), plateau $[2.8, 3.5]$ (3D) — see Papers 07–14
Γ_{inst}	instability support (activity, destabilisation)
Γ_{prod}	production support (entropy/coherence production)
Γ_{coh}	coherence support (internal consistency)
Γ_{cons}	conservation support (balance)
Γ_{corr}	correlation support (connectivity)
Γ_{int}	integrity support (structural completeness)
κ	modulation coefficient for structural free energy U_{struct}

The proxy (9) and empirical scaling (10) are not themselves the definition of the CCI. They are measurable projections of the formal structure (8) in specific contexts.

The v1.2.1 CCI diagnostics are [18]:

$$\mathrm{CCI}_{\mathrm{min}} = \frac{S}{H + B + V + \varepsilon}, \tag{11}$$

$$\mathrm{CCI}_{\mathrm{diag}} = \frac{S}{H + B + V + R_{\mathrm{rob}} + \varepsilon}. \tag{12}$$

Note: $R_{\mathrm{resp}} = (HBV)^{1/3}$ is *not* added separately — it is already encoded in H, B, V. Adding it would double-count the same information [2].

9 Dynamical Extension

The dynamical extension promotes structural response variables to local fields. The resulting pipeline [14] is:

$$d_a \;\to\; C_{ab} \;\to\; \lambda_a(t) \;\to\; \lambda_a(x,t) \;\to\; T_{\mu\nu}^{\mathrm{MAAT}} \;\to\; \mathrm{FLRW} \;\to\; \big[H(z),\, f\sigma_8(z),\, C_{\mathrm{proj}}(z)\big]. \tag{13}$$

Each arrow represents a well-defined map:

1. $d_a \to C_{ab}$: defect covariance measurement,
2. $C_{ab} \to \lambda_a(t)$: response closure Eq. (5),
3. $\lambda_a(t) \to \lambda_a(x,t)$: local field equation,
4. $\lambda_a(x,t) \to T_{\mu\nu}^{\mathrm{MAAT}}$: stress-energy coupling,
5. $T_{\mu\nu}^{\mathrm{MAAT}} \to \mathrm{FLRW}$: cosmological embedding,
6. FLRW $\to$ observables: projection layer.

Paper 43 extends this ladder one step further by embedding a bounded projection-derived coupling into an explicit linear growth equation, replacing the earlier pure growth-index proxy at the first perturbative level.

10 Observable Projection Layer

Projection observable.

$$C_{\text{proj}}(z) = \frac{S_{\text{proj}}(z)}{1 + \gamma\, D_{\text{acc}}(z)}, \tag{14}$$

where $S_{\text{proj}}(z)$ encodes expansion-driven structural breadth and $D_{\text{acc}}(z)$ encodes accessible structural depth [7].

The projection residual:

$$R_{\text{proj}}(z) = \frac{|S_{\text{proj}}(z) - \gamma D_{\text{acc}}(z)|}{S_{\text{proj}}(z) + \gamma D_{\text{acc}}(z) + \varepsilon}. \tag{15}$$

The transition estimator [15, 16]:

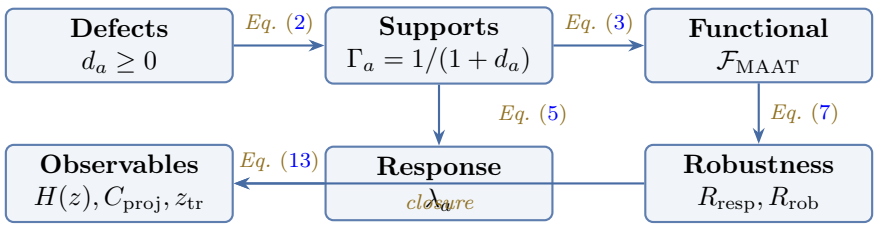
$$z_{\text{tr}} = \arg\max_z \left| \frac{d^2 R_{\text{proj}}}{dz^2} \right| \approx 1.044 \quad \text{(v0.11 baseline)}. \tag{16}$$

In MAAT v1.2.1, the three projection parameters (γ, B_*, α) are derived from response weights λ_a without free fitting [18]:

$$\gamma_\lambda = \frac{\pi_V + \pi_B}{\pi_S + \varepsilon}, \qquad B_{*,\lambda} = \frac{1}{\pi_H + \pi_B + \varepsilon}, \qquad \alpha_\lambda = 1 + \frac{\pi_S}{\pi_V + \varepsilon}. \tag{17}$$

The observable layer remains phenomenological and diagnostic. Paper 43 adds a first effective linear-growth ODE, but no Boltzmann-code implementation or full cosmological likelihood is included.

11 Architecture Diagram



12 Mathematical Status

Layer	Status	Interpretation
Defect geometry	defined	primitive diagnostics (Section 3)
Support mapping	canonical	bounded support geometry (Section 4)
Selection functional	derived	MaxEnt effective layer [1] (Section 5)
Response closure	derived	covariance-response geometry [5] (Section 6)
Robustness closure	defined	emergent compressed stability (Section 7)
CCI diagnostics	hierarchical	formal → proxy → scaling (Section 8)
Observable projection	phenomenological	diagnostic layer (Section 10)
Linear growth perturbation	first effective ODE	Paper 43 bridge, not Boltzmann-level
Boltzmann perturbation theory	open	CAMB/CLASS implementation needed
UV completion	open	unknown
Bayesian likelihood	open	future work

13 Open Problems

Principal open problems of the framework:

1. **Microscopic derivation**: the defect sectors d_a are currently defined phenomenologically. A derivation from a microscopic action or symmetry principle is missing.
2. λ_a **physical closure**: the weights are derived from defect covariance, not from fundamental physics such as entropy production rate or renormalisation group scale [4].
3. **Reference measure** μ_0: the reference ensemble that defines the unselected configuration space remains a central modelling input. A microscopic or dynamically invariant construction of μ_0 is still missing.
4. **Normalisation and cutoff structure**: the defect scales, regulator ε, activity targets, and projection cutoffs are currently operational choices. A systematic derivation or robust calibration hierarchy is still required.
5. **Perturbation theory**: Paper 43 provides a first effective linear-growth ODE. A full Boltzmann-code treatment remains open, including metric perturbations, lensing kernels, CMB anisotropies, gauge consistency, and survey-window effects.
6. **Bayesian cosmological inference**: CAMB/CLASS integration and full covariance likelihoods (BAO, CMB, WL, $H(z)$).
7. **Renormalization structure**: how do the defect sectors and weights run with energy scale?
8. **Independent validation of** $R_{\text{resp}} = (HBV)^{1/3}$: this closure should be tested on

datasets not used in calibration [12]. Here H, B, and V denote bounded support scores, not raw defects.

9. **Covariant backreaction and Bianchi consistency**: if local structural-selection fields contribute to an effective $T_{\mu\nu}^{\mathrm{MAAT}}$, the combined stress-energy sector must satisfy the appropriate covariant conservation law or an explicit exchange-current formulation.

10. **Observational constraints**: beyond toy-level projections, constraints from next-generation surveys (DESI, Euclid, Rubin).

14 Conclusion

This introductory note consolidates the mathematical architecture of the structural-selection framework developed across Papers 01–42. It provides the reader with the formal vocabulary, canonical equations, and structural hierarchy needed to follow the series.

The canonical pipeline (full series in one equation).

$$\underbrace{d_a}_{\text{defects}} \longrightarrow \underbrace{\Gamma_a}_{\text{supports}} \longrightarrow \underbrace{\mathcal{F}_{\mathrm{MAAT}}}_{\text{functional}} \longrightarrow \underbrace{\lambda_a}_{\text{response}} \longrightarrow \underbrace{R_{\mathrm{rob}}}_{\text{robustness}} \longrightarrow \underbrace{C_{\mathrm{proj}}(z)}_{\text{observable}}. \tag{18}$$

The response weights are derived from the defect covariance matrix C and target state d^*. The remaining layers are defined by canonical mappings from the primitive defect geometry. The response-closure layer reduces the number of freely chosen parameters by deriving λ_a from C and d^*; remaining modelling choices include the defect definitions, normalisations, projection ansatz, and target ensemble. No independent fifth sector is required.

The strongest result is not a claim of completeness. It is the establishment of a coherent and reproducible effective structural framework connecting nonlinear diagnostics, response geometry, and observable projection layers within a unified formal structure.

Topic	See
Defect geometry & support mapping	Papers 01–06, Paper 41
Scaling & dimensional dependence	Papers 07–14
Dynamics & control	Papers 17–20
Benchmark validation	Papers 22, 26, 27
Response closure	Papers 30, 41
Cosmological pipeline	Papers 28–35, 43
v1.2.1 robustness closure	Papers 36–39
Residual signature tests	Papers 40, 42

https://github.com/Chris4081/structural-selection-principle
https://maat-research.com

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